

Minor Project Start: **Malware Detection Tool**

Task: Start working on a basic Python script that scans a directory for suspicious files based on:

File hash comparison (using SHA-256).

Identifying executable files with suspicious behaviors.

Checking VirusTotal API (if possible).

PYTHON SCRIPT :

```
import os
```

```
import hashlib
```

```
import requests
```

```
MALWARE_HASHES = {"44d88612fea8a8f36de82e1278abb02f",  
"098f6bcd4621d373cade4e832627b4f6"} # Example malware hashes
```

```
VIRUSTOTAL_API_KEY =  
"a1f6a879863d594e34e85dbbfd349023f93c8ed45a07058bd0e889fc7bd989a3SCAN_DIRECTORY = "./test_files"
```

```
def compute_sha256(file_path):
```

```
    hasher = hashlib.sha256()
```

```
    try:
```

```
        with open(file_path, "rb") as f:
```

```
            for chunk in iter(lambda: f.read(4096), b""):
```

```
                hasher.update(chunk)
```

```
            return hasher.hexdigest()
```

```
    except Exception as e:
```

```
        print(f"Error hashing {file_path}: {e}")
```

```
    return None
```

```

def check_virustotal(file_hash):
    url = f"https://www.virustotal.com/api/v3/files/{file_hash}"
    headers = {"x-apikey": VIRUSTOTAL_API_KEY}
    try:
        response = requests.get(url, headers=headers)
        if response.status_code == 200:
            result = response.json()
            positives = result.get("data", {}).get("attributes", {}).get("last_analysis_stats",
            {}).get("malicious", 0)
            return positives > 0
        except Exception as e:
            print("Error contacting VirusTotal:", e)
    return False

```

```

def scan_directory():
    print(f"Scanning directory: {SCAN_DIRECTORY}\n")
    for root, _, files in os.walk(SCAN_DIRECTORY):
        for file in files:
            file_path = os.path.join(root, file)
            file_hash = compute_sha256(file_path)
            if file_hash:
                print(f"[CHECKING] {file} => {file_hash}")

            if file_hash in MALWARE_HASHES:
                print(f"[WARNING] Malicious hash detected: {file}")

            if file.endswith((".exe", ".dll", ".bin", ".sh", ".out")):
                print(f"[ALERT] Suspicious executable found: {file}")

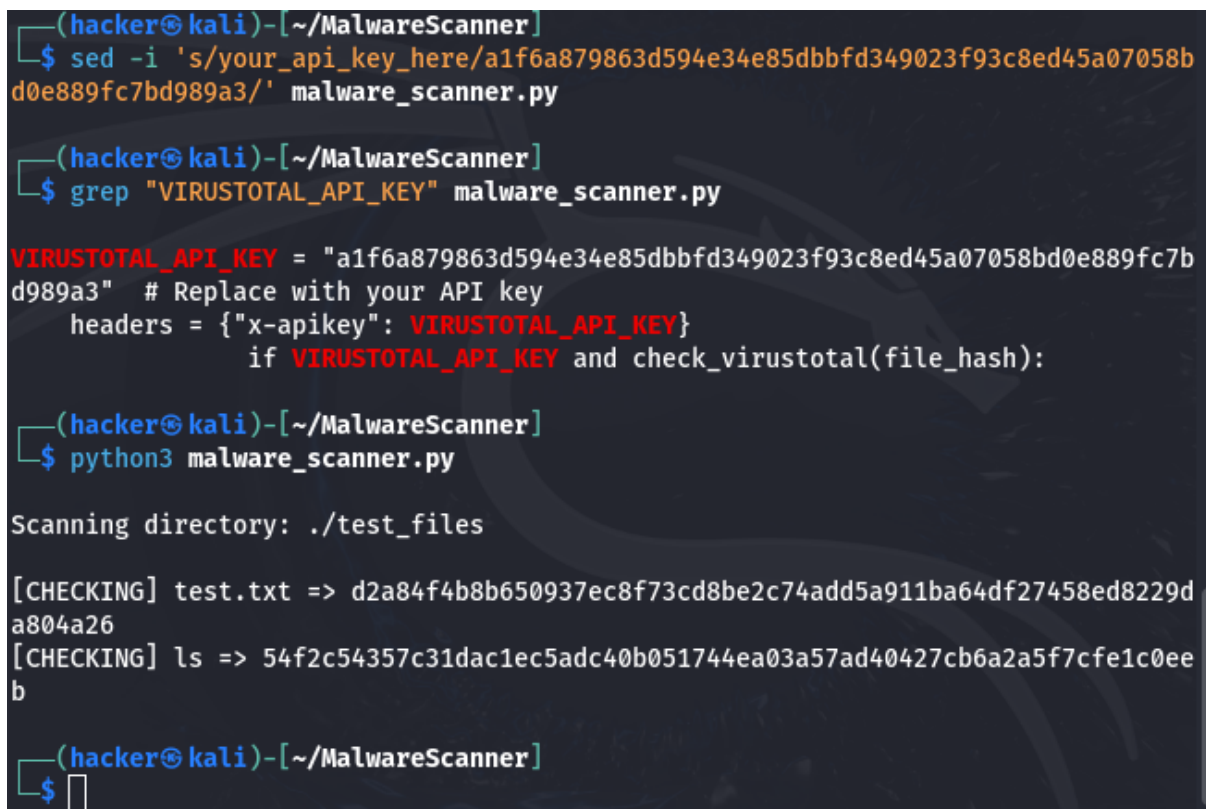
```

```
if VIRUSTOTAL_API_KEY and check_virustotal(file_hash):  
    print(f"[DANGER] {file} flagged as malicious by VirusTotal!")
```

Run the scan

```
if __name__ == "__main__":  
    scan_directory()
```

Screenshots :



```
(hacker@kali)-[~/MalwareScanner]  
$ sed -i 's/your_api_key_here/a1f6a879863d594e34e85dbbfd349023f93c8ed45a07058bd0e889fc7bd989a3/' malware_scanner.py  
  
(hacker@kali)-[~/MalwareScanner]  
$ grep "VIRUSTOTAL_API_KEY" malware_scanner.py  
  
VIRUSTOTAL_API_KEY = "a1f6a879863d594e34e85dbbfd349023f93c8ed45a07058bd0e889fc7bd989a3" # Replace with your API key  
headers = {"x-apikey": VIRUSTOTAL_API_KEY}  
    if VIRUSTOTAL_API_KEY and check_virustotal(file_hash):  
  
(hacker@kali)-[~/MalwareScanner]  
$ python3 malware_scanner.py  
  
Scanning directory: ./test_files  
  
[CHECKING] test.txt => d2a84f4b8b650937ec8f73cd8be2c74add5a911ba64df27458ed8229da804a26  
[CHECKING] ls => 54f2c54357c31dac1ec5adc40b051744ea03a57ad40427cb6a2a5f7cfe1c0eeb  
  
(hacker@kali)-[~/MalwareScanner]  
$
```